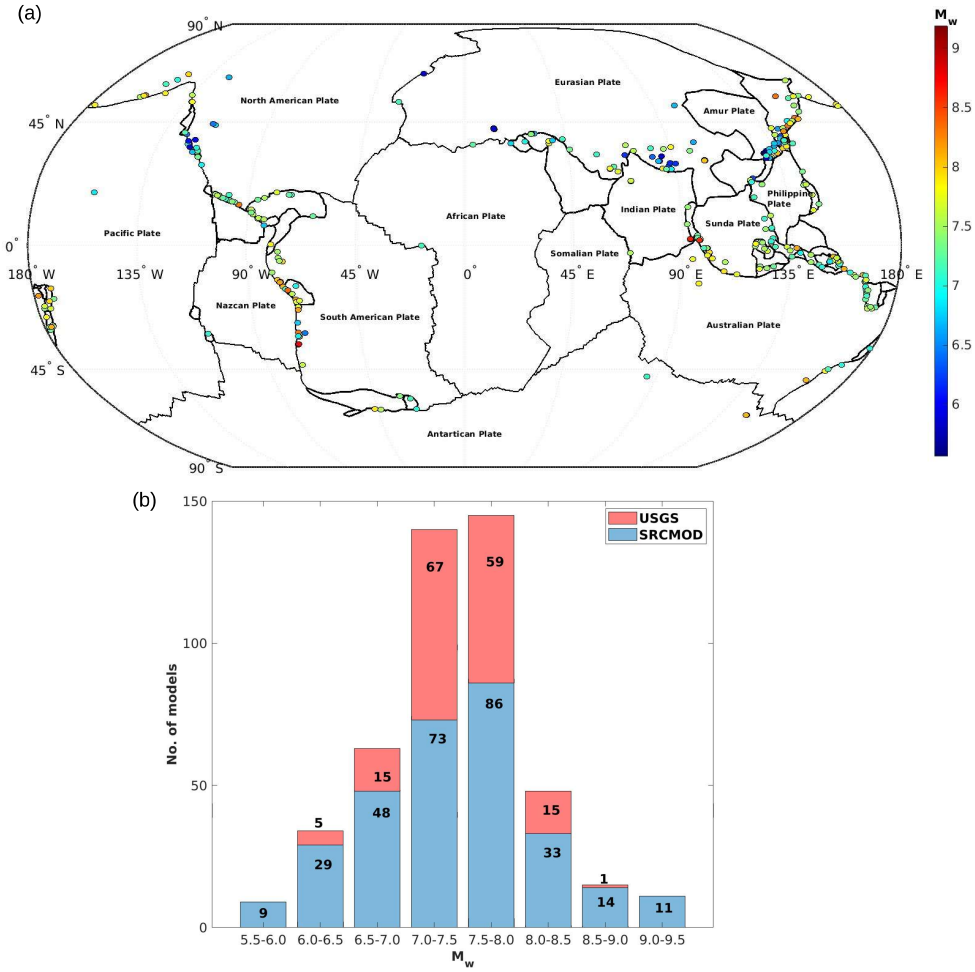

Figure 1: Distribution of 465 rupture models considered in this study: (a) Distribution of the location with plate boundaries. (b) Histogram showing the distribution of the events with magnitude.

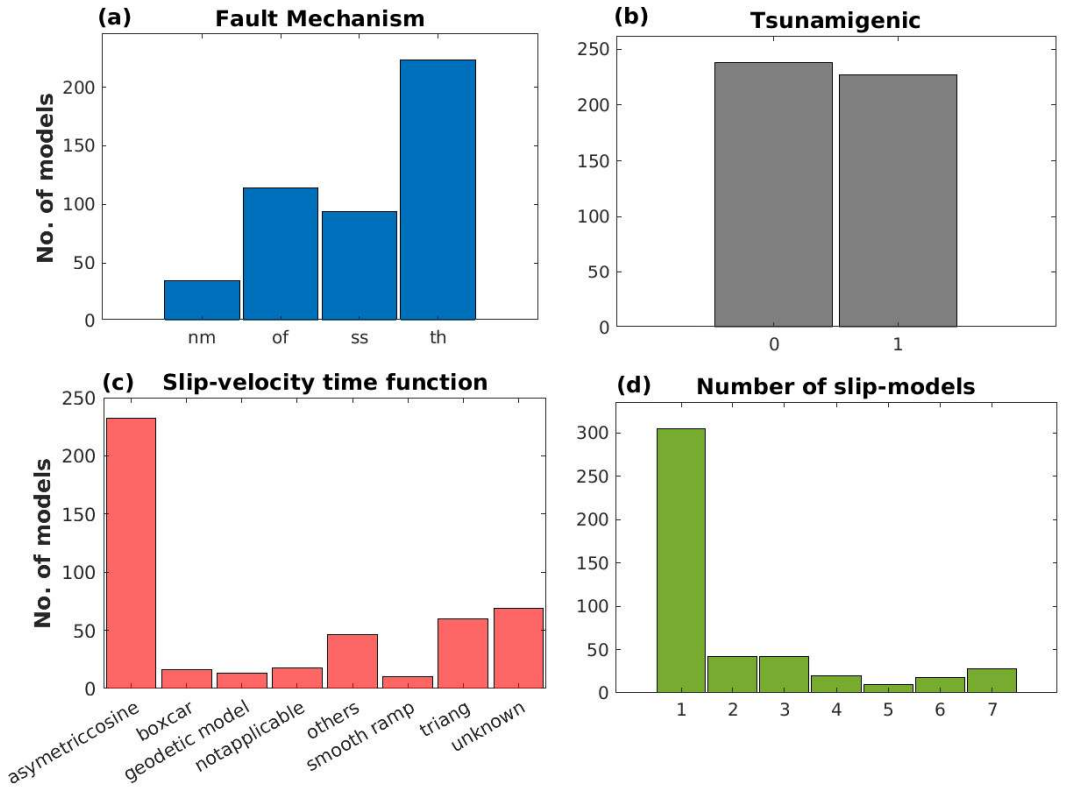

Figure 2: Distribution of the source models into various random effect levels: (a) Faulting mechanism type (FM), (b) Tsunamigenic and non-tsunamigenic events (TS), (c) Slip-velocity time function (SVT) and (d) number of slip models (NM). Note: The SVT tags are based on the information from SRCMOD and USGS databases. For tags having less than 10 population, they are clubbed into the others tags.

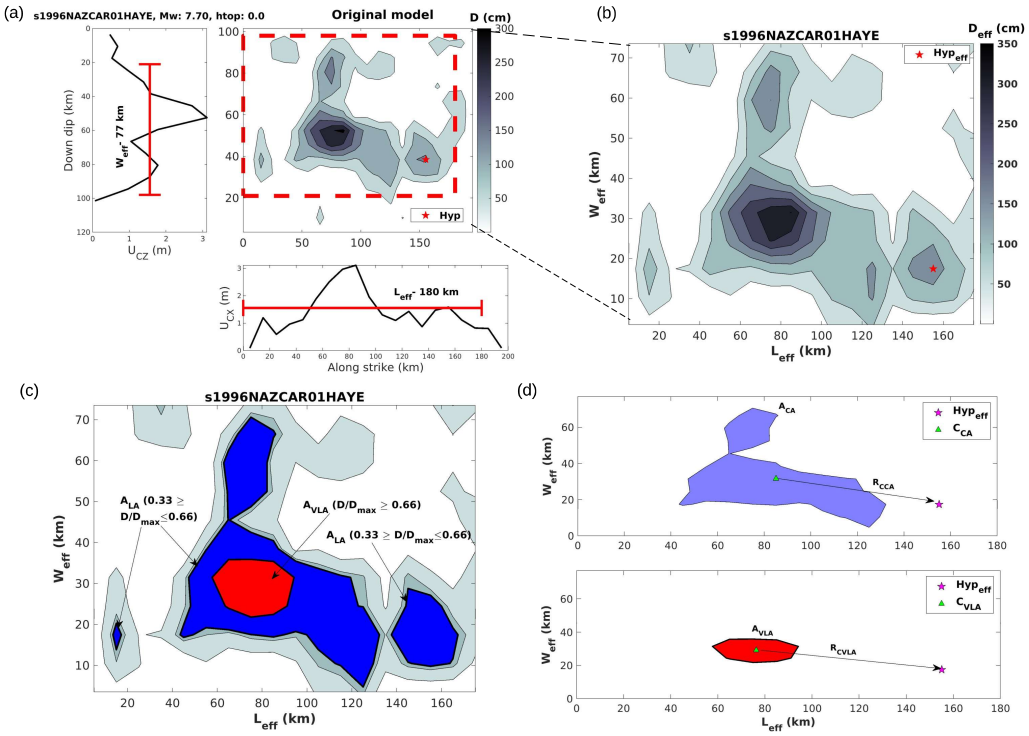

Figure 3: Estimation of first-order scaling parameters for the 1996 Nazcar earthquake event: (a) Source dimension from inversion model, (b) Effective slip field, (c) Regions of slip-asperity in the rupture model, (d) Distance parameters with respect to the centroid.

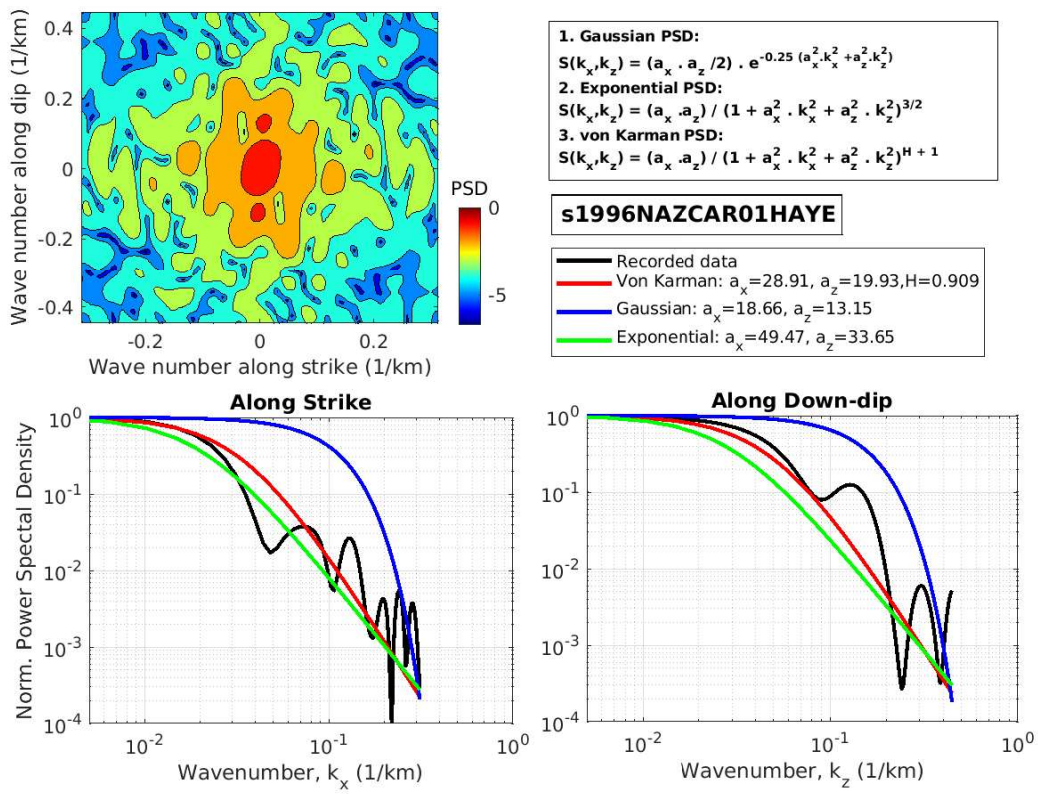

Figure 4: Estimation of second-order scaling parameters for 1996 Nazcar earthquake event. The correlation lengths are estimated using spectral analysis of the slip data. The bottom plots show the comparison between the Gaussian, exponential and von Karman power spectral density (PSD) at the cross-sections along the strike $(k_x) = 0$ and along the dip$(k_z) = 0$. Note: a_x and a_z are the correlation lengths along the strike and along the dip, respectively.

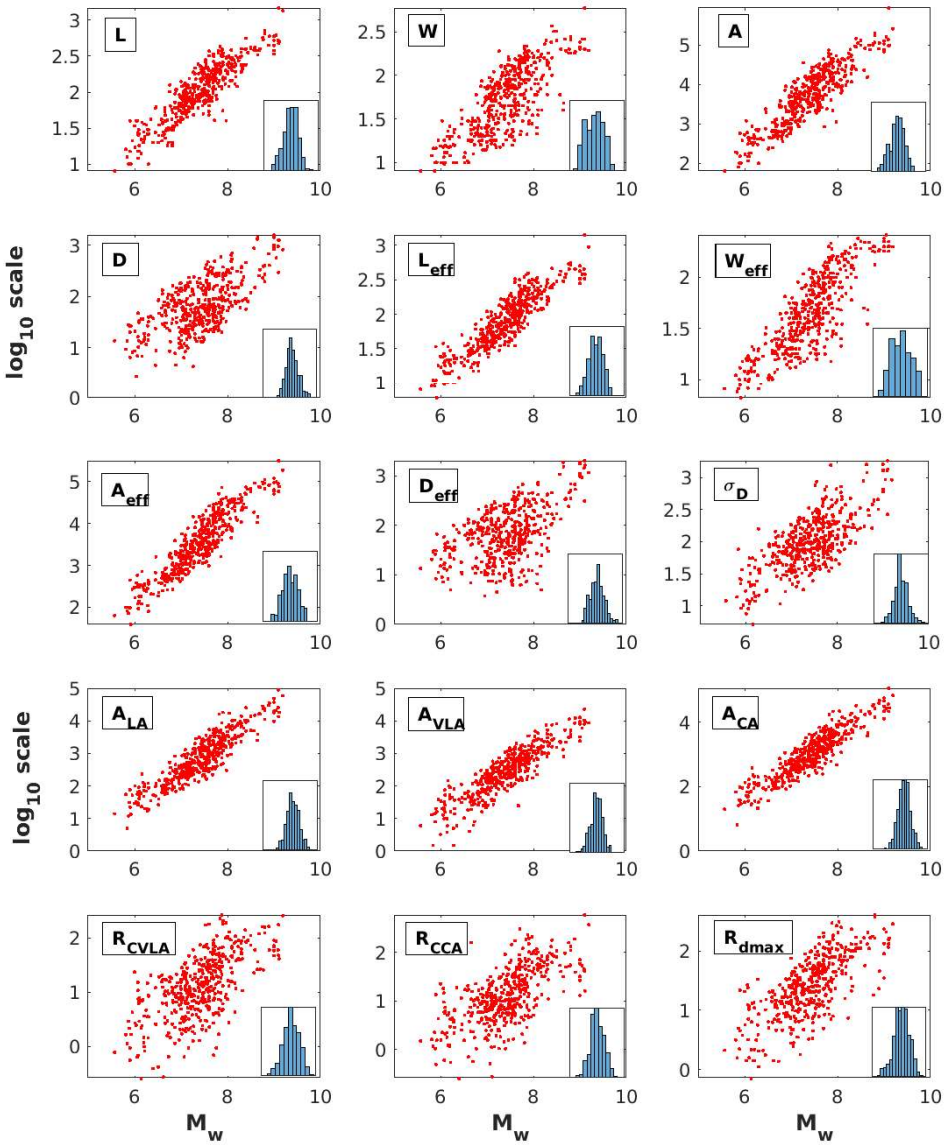

Figure 5: Scatter plots showing the distribution of the scaling parameters with earth magnitude (M_w). The plots show that the parameters are distributed linearly with M_w in log_{10} scale. The figure also includes histogram plots for the log_{10} scaling parameters values.

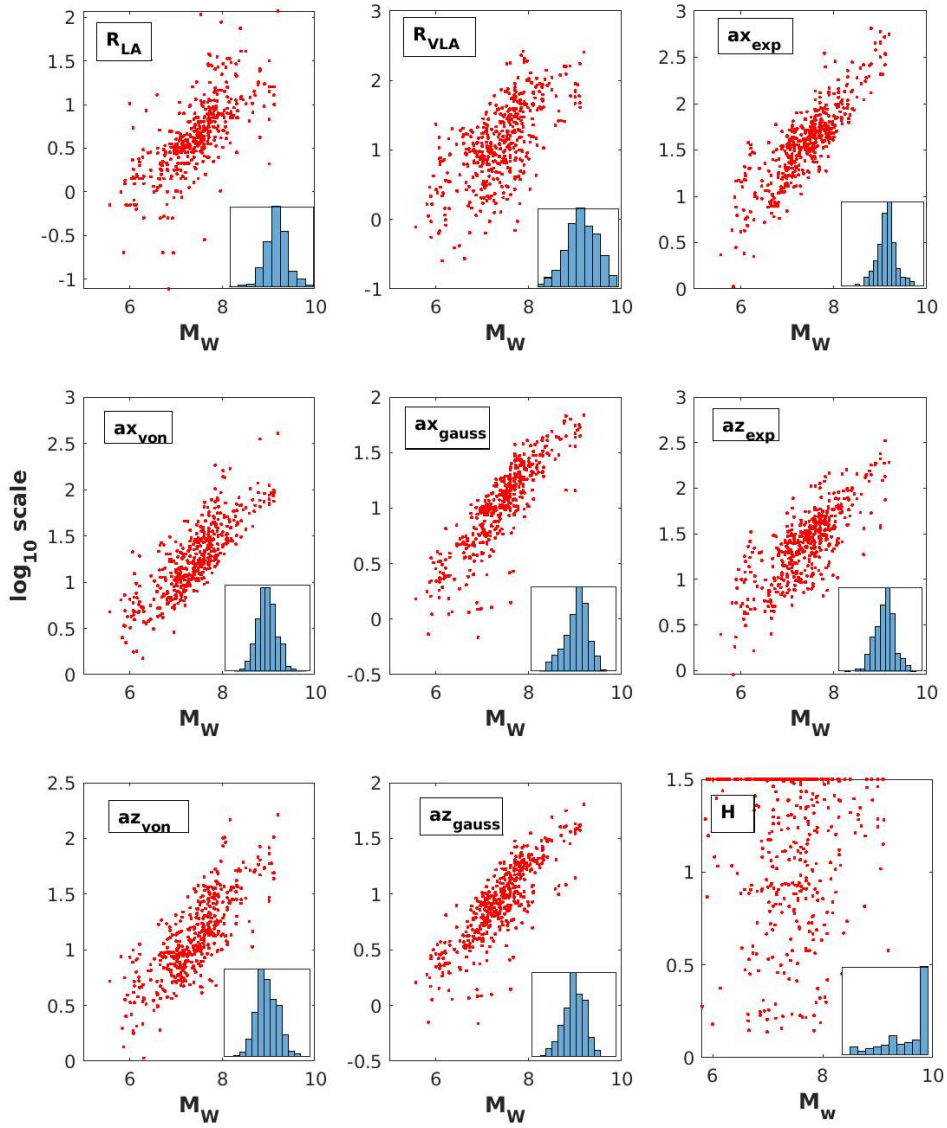

Figure 5: (Contd..) The plots show that the parameters are distributed linearly with M_w in log_{10} scale except for the Hurst exponent (H). The figure also includes histogram plots for the log_{10} scaling parameters values. Note: The plot of the Hurst exponent is on a normal scale and has not been included in the scaling relationship.

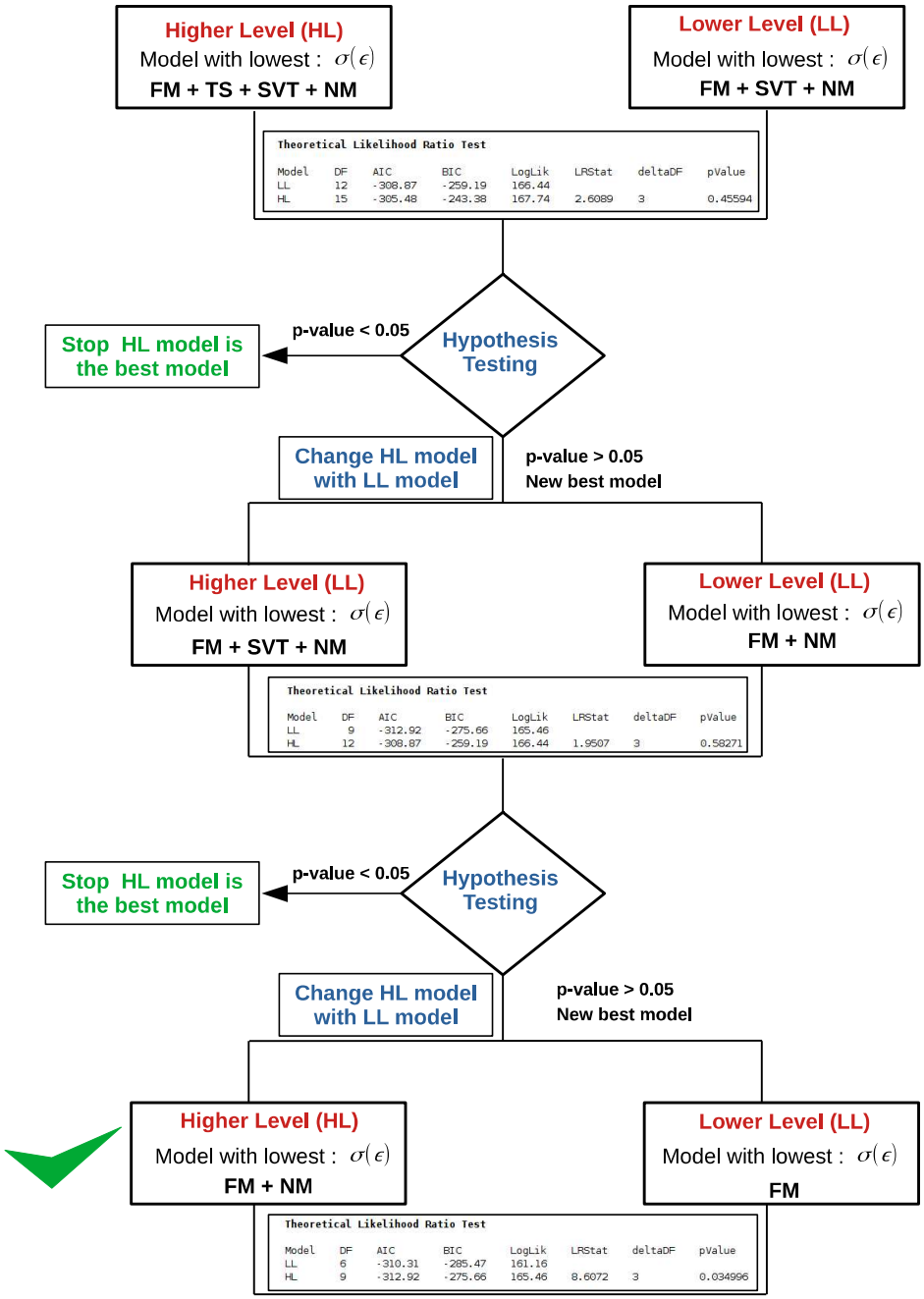

Figure 6: Flowchart demonstrating the selection of the best mixed-effect model for effective length (L_{eff}). The higher-level and lower-level models change each time if the p-value > 0.05. A similar procedure is repeated for other scaling parameters.

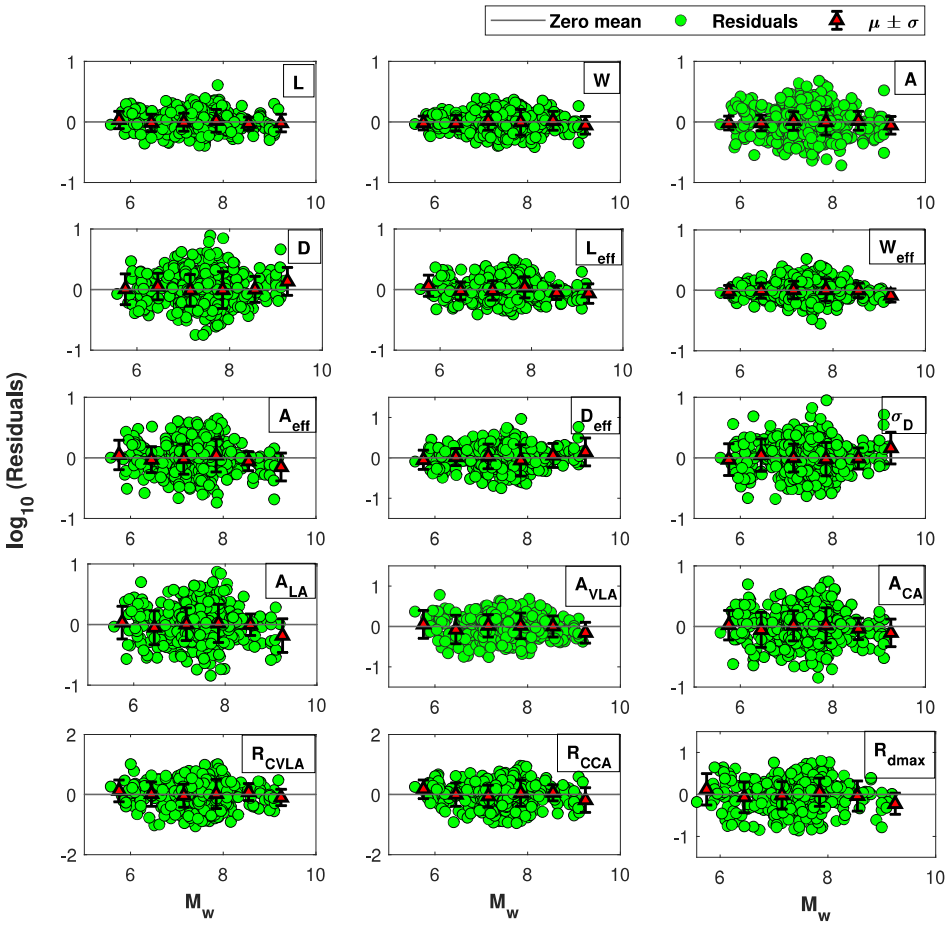

Figure 7: Residual plots showing the distribution of residuals with respect to magnitude around the zero-mean line. The plots also include error-plots for mean residuals and standard deviation.

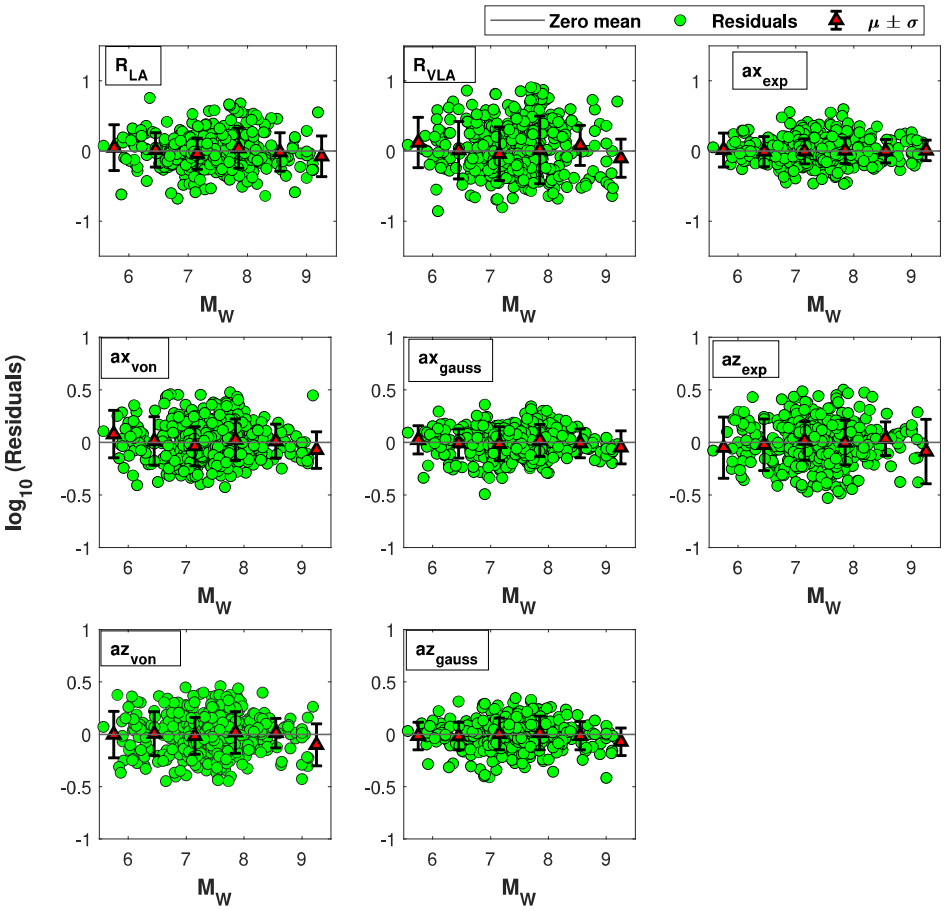

Figure 7: (Contd..) Residual plots showing the distribution of residuals with respect to magnitude around the zero-mean line. The plots also include error-plots for mean residuals and standard deviation.

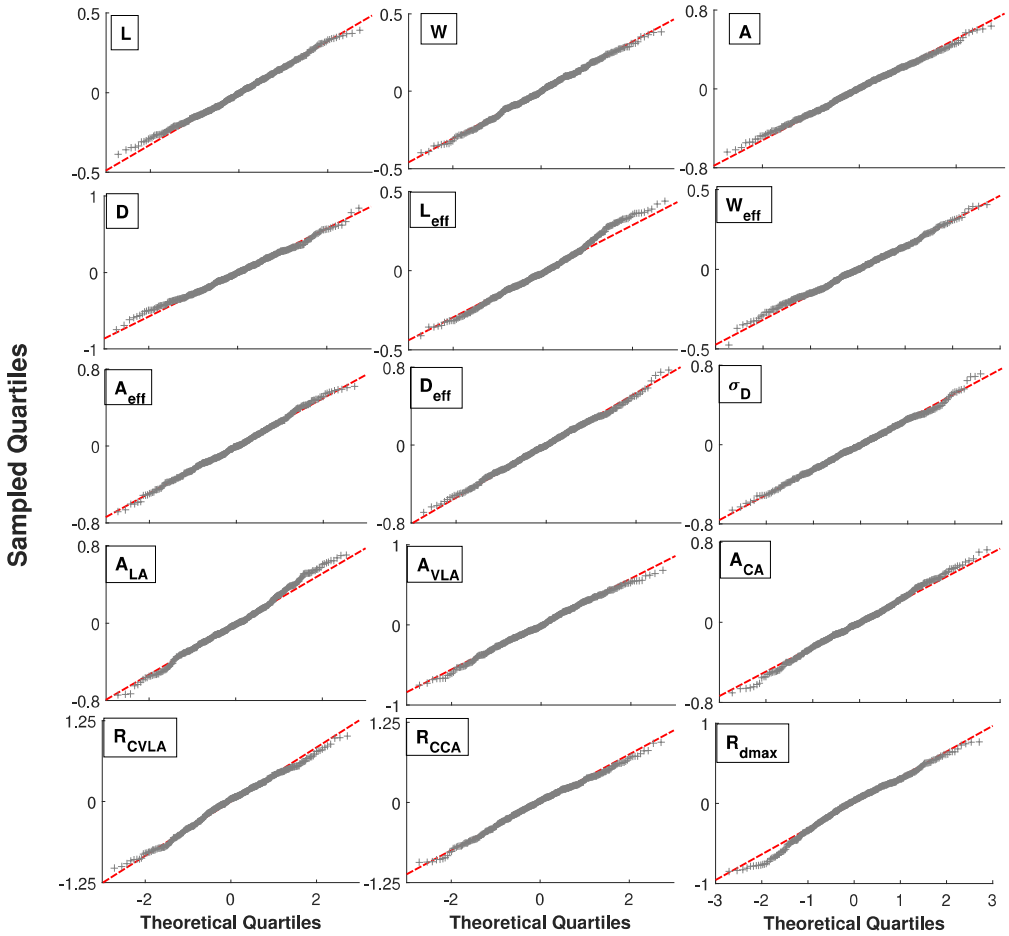

Figure 8: Q-Q plots of residuals for different scaling parameters. For an ideal case, the residuals should lie on the red line. The plots verify the normality of the residuals.

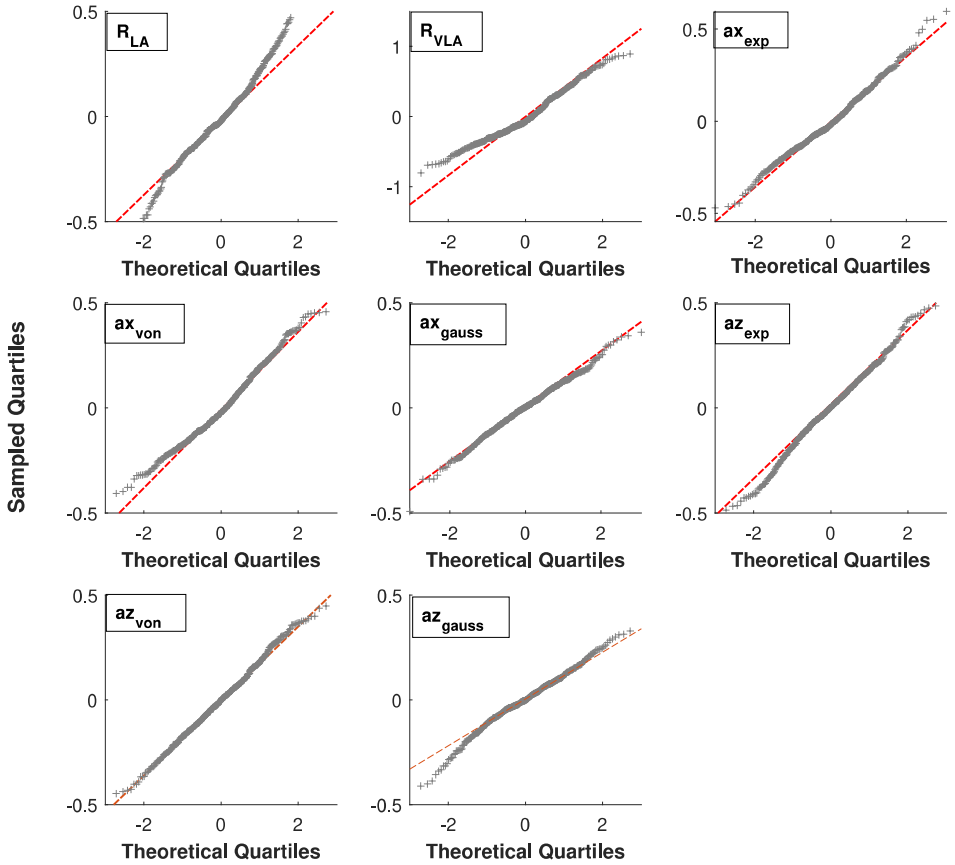

Figure 8: (Contd..) Q-Q plots of residuals for different scaling parameters. For an ideal case, the residuals should lie on the red line. The plots verify the normality of the residuals.

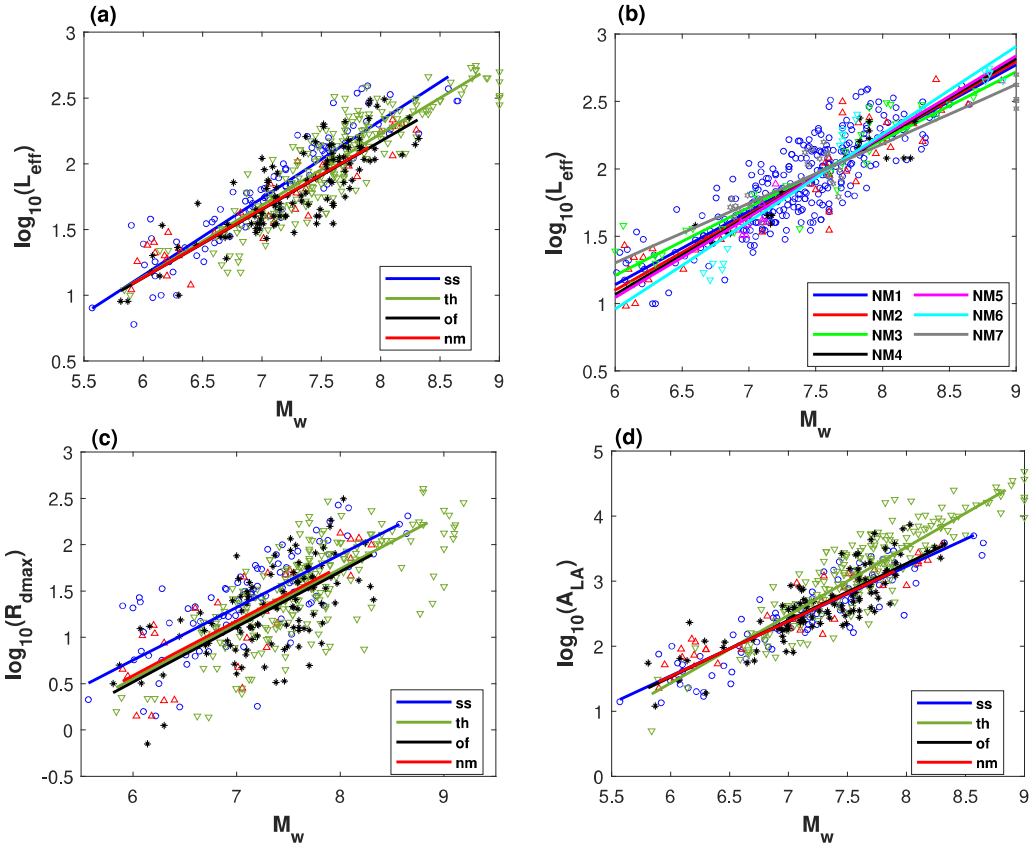

Figure 9: Scaling variation plot for first-order scaling parameter against magnitude (M_w): (a) effective length (L_{eff}) for different faulting type, here the number of slip models (NM) is taken as 1 corresponding to highest population. (b) L_{eff} for different number of slip models, here the faulting type is taken as thrust fault (th). (c) minimum distance of the hypocentre fro point of max-slip (R_{dmax}) for different faulting types. (d) Area of large asperity (A_{LA}) for different faulting types. Note: for the corresponding functional form corresponding to each scaling parameter, please refer to Table 2.

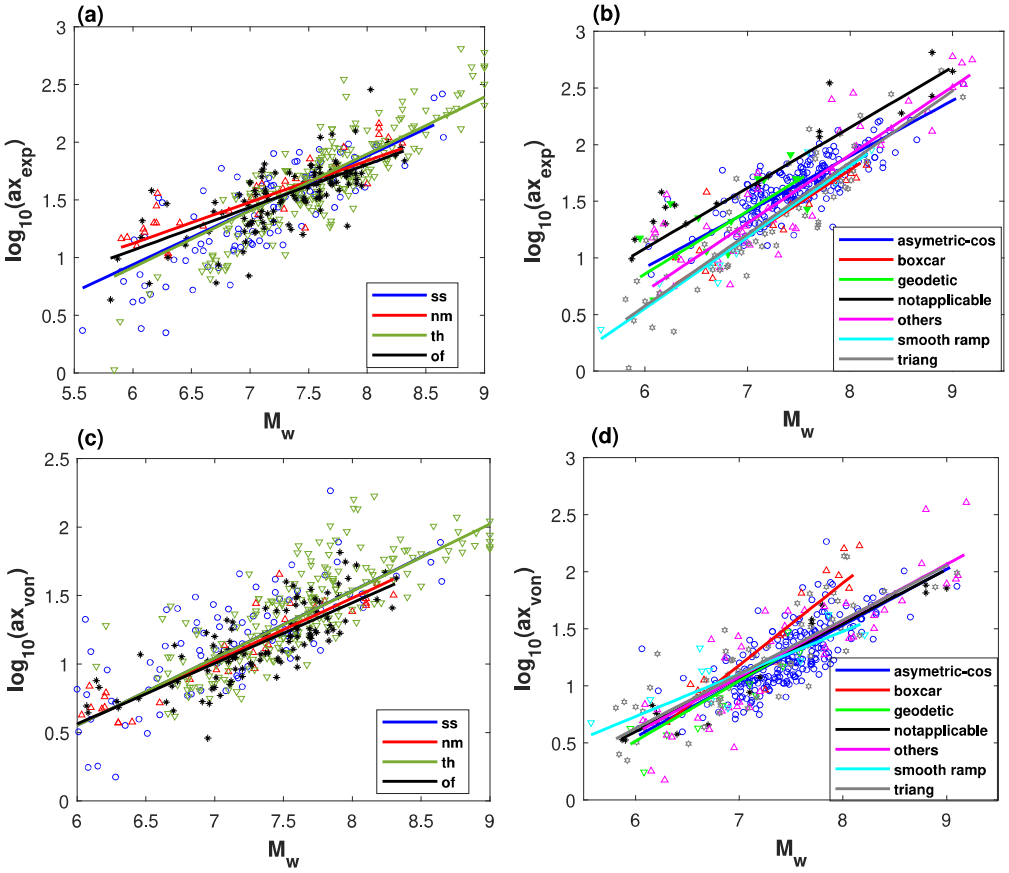

Figure 10: Scaling variation plot for second-order scaling parameter against magnitude (M_w): (a) ax_{exp} vs FM, here the SVT is asymetriccosine corresponding to the highest population, (b) axe_{exp} vs SVT, here, the FM is considered as thrust faulting (th), (c) ax_{von} vs FM and SVT is asymetriccosine, (d) ax_{von} vs SVT, FM is taken as thrust faulting (th). Note: for the corresponding functional form corresponding to each scaling parameter, please refer to Table 2.

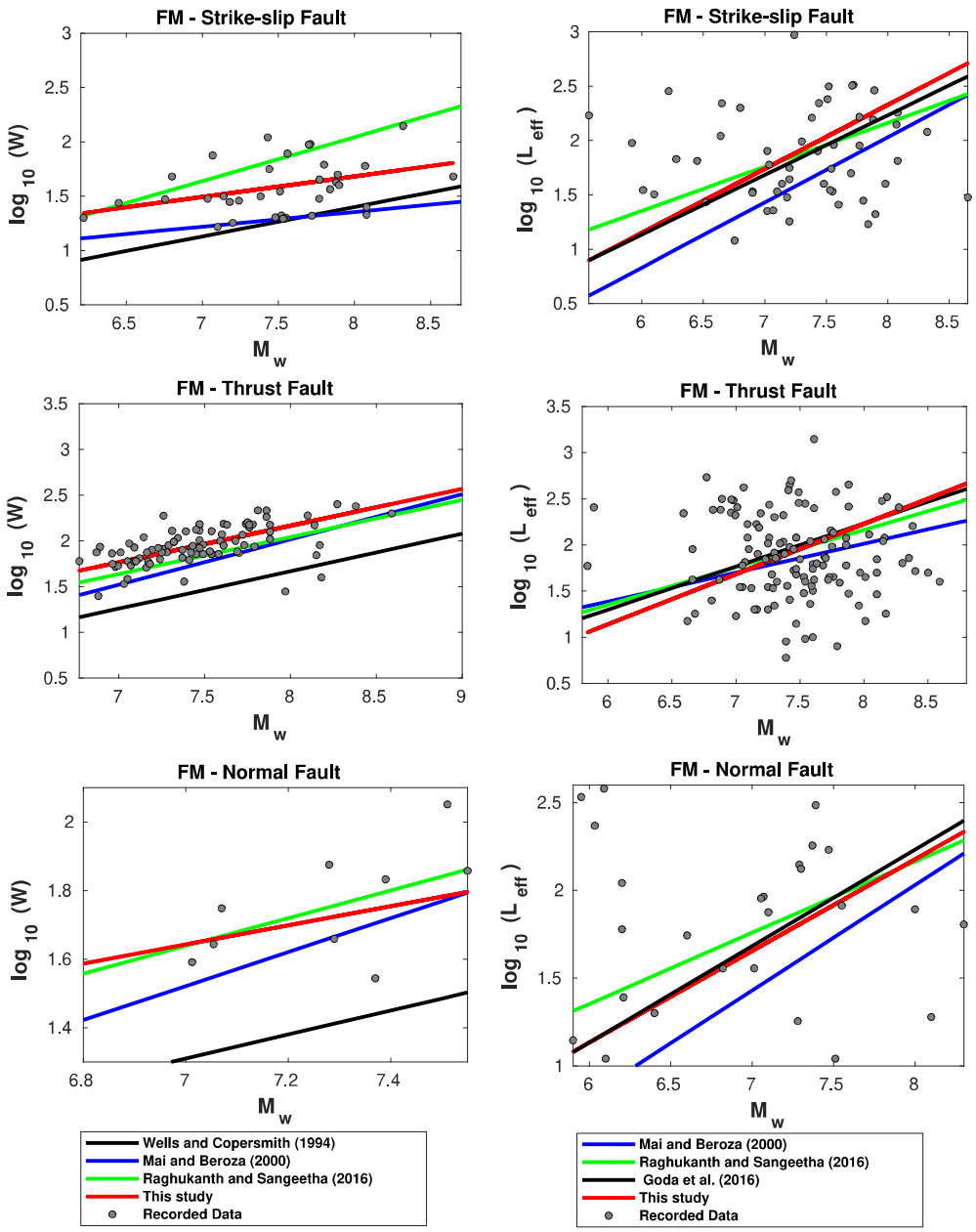

Figure 11: Comparison plots between the developed prediction models and scaling relationships from other studies with the recorded data. The left side of the plot contains the comparison of W for different FM. For the predictive model of W, the SVT is taken as asymmetric-cosine, and NM is taken as 1. The right side of the plot contains the comparison for L_{eff} for different FM, for which NM is taken as 1.

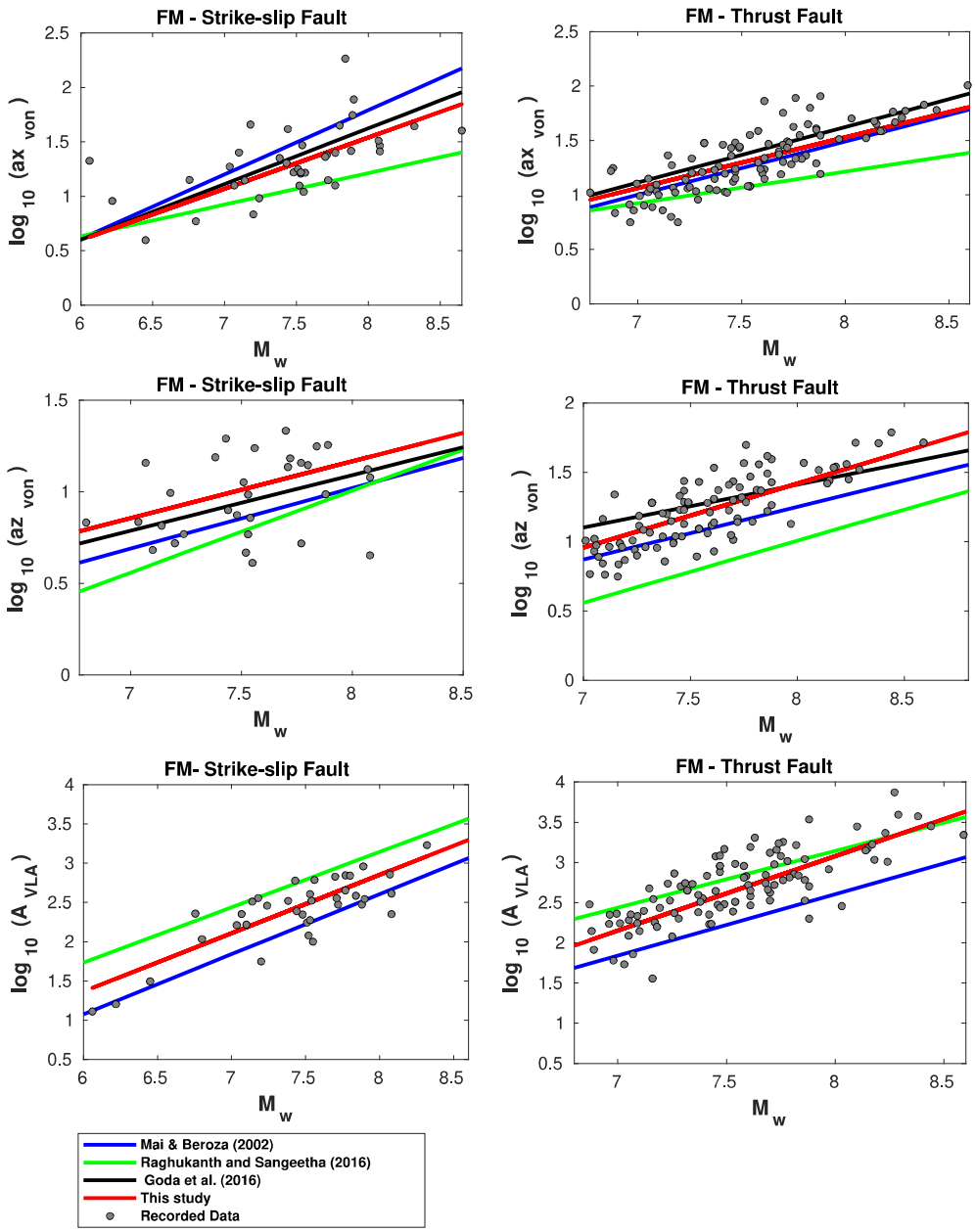

Figure 12: Comparison plots between the developed prediction models and scaling relationships from other studies with the recorded data. The plots at the top and middle represent scaling relationships for axe_{von} and az_{von}, respectively. The bottom two plots are for scaling of A_{VLA} for different FM types. In all the plots, the SVT is taken as asymmetric-cosine.

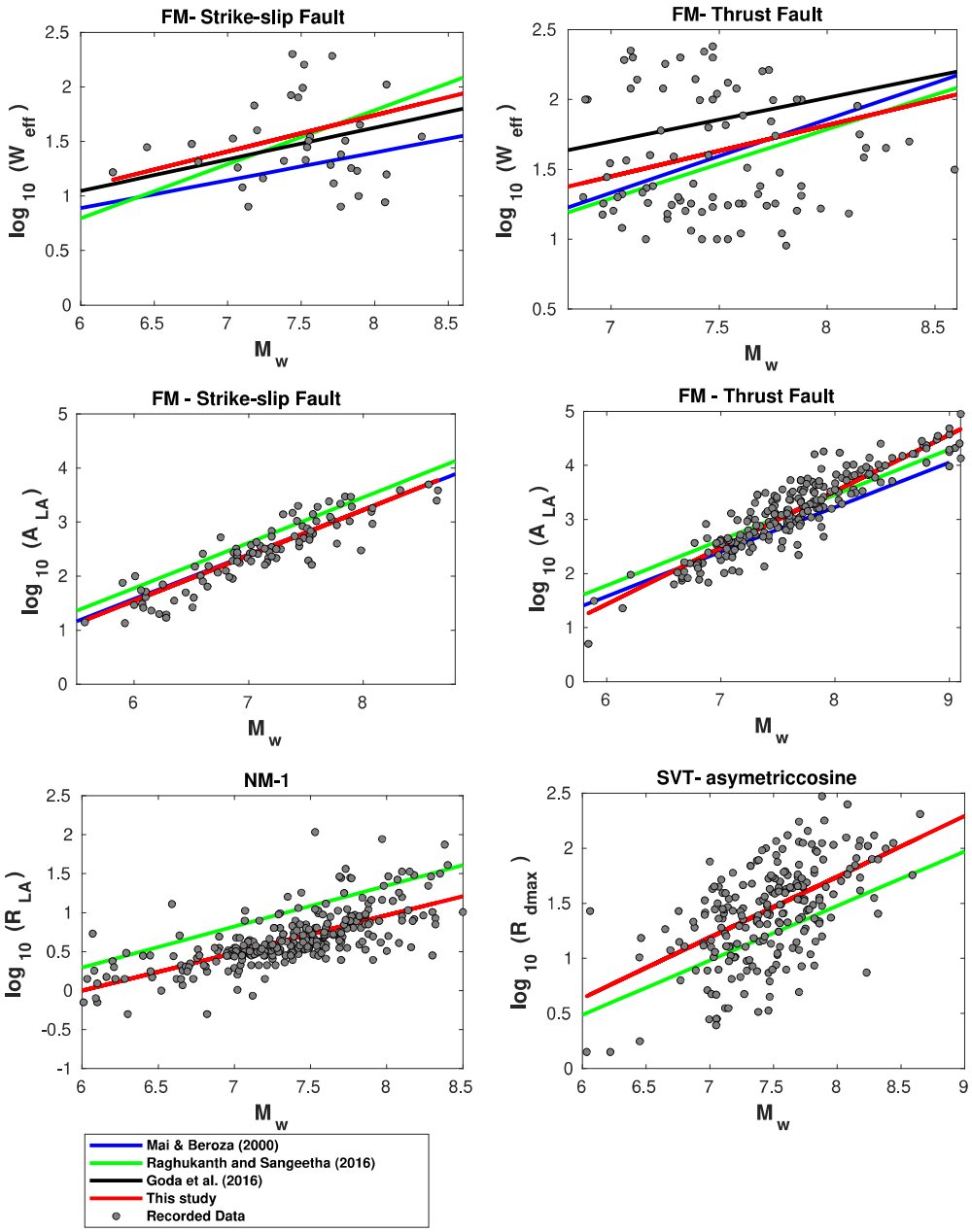

Figure 13: Comparison plots between the prediction models and scaling relationships from other studies with the recorded data. For W_{eff}, SVT is taken as asymmetric-cosine, and NM is taken as 1. For thrust fault, TS is taken as 1; otherwise, 0. For parameters R_{LA} and R_{dmax}, NM and SVT are 1 and asymmetric-cosine, respectively.

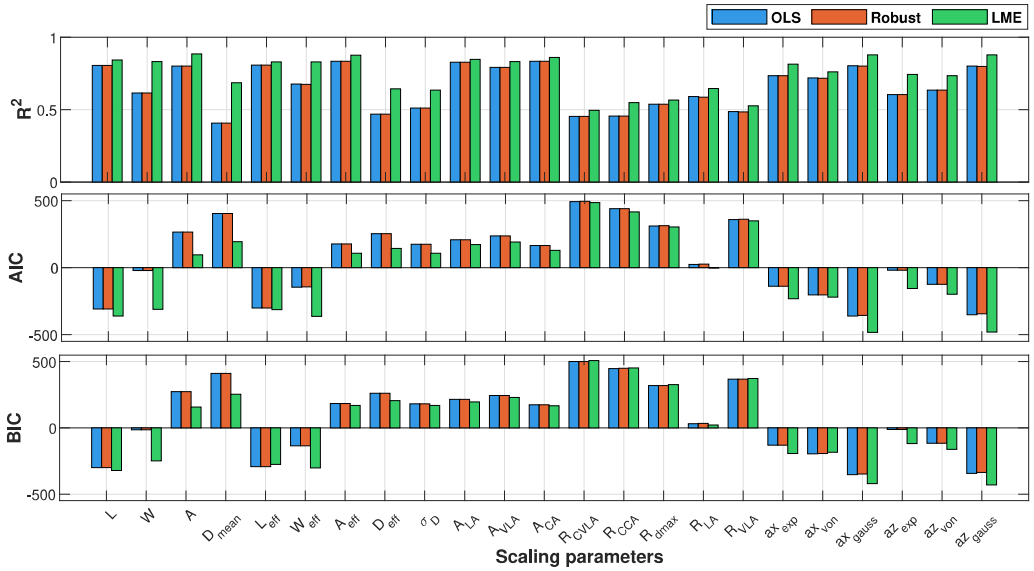

Figure 14: Comparison between ordinary least squares (OLS), Robust, and mixed effect (LME) linear regression models for different scaling parameters.